1) a)

$3\left(3^{k-1}+\frac{1}{2}\right)-1=3\times3^{k-1}+\frac{3}{2}-1=3^1\times3^{k-1}+\frac{1}{2}=3^k+\frac{1}{2}$ as req.

1) b)

$3(2^k+3^k)-2^k=3\times2^k+3\times3^k-2^k=2^1\times2^k+3^1\times3^k=2^{k+1}+3^{k+1}$ as req.

1) c)

$\frac{\frac{3}{3k-1}}{1+\frac{3}{3k-1}}=\frac{3}{3k-1+3}=\frac{3}{3k+2}=\frac{3}{3(k+1)-1}$ as req.

1) d)

$\frac{1}{2}(k^2-5k+12)+k-2=\frac{1}{2}(k^2-5k+12+2k-4)$

$=\frac{1}{2}(k^2-3k+8)$

$=\frac{1}{2}((k+1)^2-5k+7)$

$=\frac{1}{2}((k+1)^2-5(k+1)+12)$ as req.

2)

Prove true for $n=1$:

$u_1=11-3\times1=8$ so true for $n=1$

Assume true for $n=k$:

$u_k=11-3k$

Prove true for $n=k+1$:

$u_{k+1}=u_k-3$

$=11-3k-3$

$=11-3(k+1)$

which is $u_n=11-3n$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq1$

3)

Prove true for $n = 1$:

$u_1 = 2^{1-1} + 1 = 1 + 1 = 2$ so true for $n = 1$

Assume true for $n = k$:

$u_k = 2^{k-1} + 1$

Prove true for $n = k + 1$:

$u_{k+1} = 2u_k - 1$

$= 2(2^{k-1} + 1) - 1$

$= 2 \times 2^{k-1} + 2 - 1$

$= 2^1 \times 2^{k-1} + 2 - 1$

$= 2^k + 1$

$= 2^{(k+1)-1} + 1$

which is $u_n = 2^{n-1} + 1$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

4)

Prove true for $n = 1$:

$u_1 = 9 \times 4^{1-1} - 1 = 9 - 1 = 8$ so true for $n = 1$

Assume true for $n = k$:

$u_k = 9 \times 4^{k-1} - 1$

Prove true for $n = k + 1$:

$u_{k+1} = 4u_k + 3$

$= 4(9 \times 4^{k-1} - 1) + 3$

$= 9 \times 4^1 \times 4^{k-1} - 4 + 3$

$= 9 \times 4^k - 1$

$= 9 \times 4^{(k+1)-1} - 1$

which is $u_n = 9 \times 4^{n-1} - 1$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

5)

Prove true for $n = 1$:

$u_1 = \frac{1}{6}(7 \times 3^1 + 3) = \frac{1}{6}(21 + 3) = 4$ so true for $n = 1$

Assume true for $n = k$:

$u_k = \frac{1}{6}(7 \times 3^k + 3)$

Prove true for $n = k + 1$:

$u_{k+1} = 3u_k - 1$

$= 3\left[\frac{1}{6}(7 \times 3^k + 3)\right] - 1$

$= \frac{1}{6}(7 \times 3^1 \times 3^k + 9) - 1$

$= \frac{1}{6}(7 \times 3^1 \times 3^k + 9 - 6)$

$= \frac{1}{6}(7 \times 3^{k+1} + 3)$

which is $u_n = \frac{1}{6}(7 \times 3^n + 3)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

6)

Prove true for $n = 1$:

$u_1 = \frac{19\times8-40}{56} = \frac{112}{56} = 2$ so true for $n = 1$

Assume true for $n = k$:

$$u_k = \frac{19 \times 8^k - 40}{56}$$

Prove true for $n = k + 1$:

$u_{k+1} = 8u_k + 5$

$= 8\left(\frac{19\times8^k-40}{56}\right) + 5$

$= \frac{19\times8\times8^k-320}{56} + \frac{280}{56}$

$= \frac{19\times8^1\times8^k-40}{56}$

$= \frac{19\times8^{k+1}-40}{56}$

which is $u_n = \frac{19\times8^n-40}{56}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

7)

Prove true for $n = 1$:

$u_1 = \frac{1}{2}(3 + 7 - 6) = \frac{1}{2} \times 4 = 2$ so true for $n = 1$

Assume true for $n = k$:

$u_k = \frac{1}{2}(3k^2 + 7k - 6)$

Prove true for $n = k + 1$:

$u_{k+1} = u_k + 3k + 5$

$= \frac{1}{2}(3k^2 + 7k - 6) + 3k + 5$

$= \frac{1}{2}(3k^2 + 7k - 6 + 6k + 10)$

$= \frac{1}{2}(3k^2 + 13k + 4)$

$= \frac{1}{2}(3(k + 1)^2 + 7k + 1)$

$= \frac{1}{2}(3(k + 1)^2 + 7(k + 1) - 6)$

which is $u_n = \frac{1}{2}(3n^2 + 7n - 6)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8)

Prove true for $n = 1$:

$u_1 = \frac{1}{1} = 1$ so true for $n = 1$

Assume true for $n = k$:

$u_k = \frac{1}{k}$

Prove true for $n = k + 1$:

$u_{k+1} = \frac{u_k}{1 + u_k}$

$= \frac{\frac{1}{k}}{1+\frac{1}{k}}$

$= \frac{1}{k+1}$

which is $u_n = \frac{1}{n}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9)

Prove true for $n = 1$:

$u_1 = \frac{3^1+1}{3^1-1} = \frac{4}{2} = 2$ so true for $n = 1$

Assume true for $n = k$:

$u_k = \frac{3^k+1}{3^k-1}$

Prove true for $n = k + 1$:

$u_{k+1} = \frac{2u_k+1}{u_k+2}$

$= \frac{2\left(\frac{3^k+1}{3^k-1}\right)+1}{\left(\frac{3^k+1}{3^k-1}\right)+2}$

$= \frac{2(3^k+1)+3^k-1}{3^k+1+2(3^k-1)}$

$= \frac{2\times3^k+2+3^k-1}{3^k+1+2\times3^k-2}$

$= \frac{3\times3^k+1}{3\times3^k-1}$

$= \frac{3^{k+1}+1}{3^{k+1}-1}$

which is $u_n = \frac{3^n+1}{3^n-1}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

10) a)

$z_2 = (1 + \mathrm{i})z_1 + (1 - \mathrm{i})^1 = (1 + \mathrm{i})(1 + \mathrm{i}) + (1 - \mathrm{i}) = 1 + 2\mathrm{i} - 1 + 1 - \mathrm{i} = 1 + \mathrm{i}$

$z_3 = (1 + \mathrm{i})z_2 = (1 + \mathrm{i})(1 + \mathrm{i}) + (1 - \mathrm{i})^2 = 2\mathrm{i} + (1 - 2\mathrm{i} - 1) = 0$

10) b)

Prove true for $n = 1$:

$z_1 = \frac{1}{2}(1+\mathrm{i})^1 + \frac{\mathrm{i}}{2}(1-\mathrm{i})^1 = \frac{1}{2} + \frac{1}{2}\mathrm{i} + \frac{1}{2}\mathrm{i} + \frac{1}{2} = 1 + \mathrm{i}$

so true for $n = 1$

Assume true for $n = k$:

$$z_k = \frac{1}{2}(1+\mathrm{i})^k + \frac{\mathrm{i}}{2}(1-\mathrm{i})^k$$

Prove true for $n = k + 1$:

$z_{k+1} = (1+\mathrm{i})z_k + (1-\mathrm{i})^k$

$= (1+\mathrm{i})\left[\frac{1}{2}(1+\mathrm{i})^k + \frac{\mathrm{i}}{2}(1-\mathrm{i})^k\right] + (1-\mathrm{i})^k$

$$= \frac{1}{2}(1+\mathrm{i})(1+\mathrm{i})^k + \frac{\mathrm{i}}{2}(1+\mathrm{i})(1-\mathrm{i})^k + (1-\mathrm{i})^k$$

$$= \frac{1}{2}(1+\mathrm{i})^{k+1} + \frac{\mathrm{i}}{2}(1-\mathrm{i})^k\left[(1+\mathrm{i}) + \frac{2}{\mathrm{i}}\right]$$

$$= \frac{1}{2}(1+\mathrm{i})^{k+1} + \frac{\mathrm{i}}{2}(1-\mathrm{i})^k[1-\mathrm{i}]$$

$= \frac{1}{2}(1+\mathrm{i})^{k+1} + \frac{\mathrm{i}}{2}(1-\mathrm{i})^{k+1}$

which is $z_n = \frac{1}{2}(1+\mathrm{i})^n + \frac{\mathrm{i}}{2}(1-\mathrm{i})^n$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

11) a)

$u_2 = \frac{u_1}{2+u_1} = \frac{3}{2+3} = \frac{3}{5}$

$u_3 = \frac{u_2}{2+u_2} = \frac{\frac{3}{5}}{2+\frac{3}{5}} = \frac{3}{13}$

$u_4 = \frac{u_3}{2+u_3} = \frac{\frac{3}{13}}{2+\frac{3}{13}} = \frac{3}{29}$

11) b)

$2^{1+1} - 3 = 1$

$2^{2+1} - 3 = 5$

$2^{3+1} - 3 = 13$

$2^{4+1} - 3 = 29$

11) c)

$u_n = \frac{3}{2^{n+1}-3}$

11) d)

Prove true for $n = 1$:

$u_1 = \frac{3}{2^{1+1}-3} = \frac{3}{4-3} = 3$ so true for $n = 1$

Assume true for $n = k$:

$u_k = \frac{3}{2^{k+1}-3}$

Prove true for $n = k + 1$:

$u_{k+1} = \frac{u_k}{2+u_k}$

$= \frac{\frac{3}{2^{k+1}-3}}{2+\frac{3}{2^{k+1}-3}}$

$= \frac{3}{2(2^{k+1}-3)+3}$

$= \frac{3}{2\times 2^{k+1}-6+3}$

$= \frac{3}{2^1\times 2^{k+1}-3}$

$= \frac{3}{2^{k+2}-3}$

$= \frac{3}{2^{(k+1)+1}-3}$

which is $u_n = \frac{3}{2^{k+1}-3}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

12)

Prove true for $n = 1$:

$u_1 = q + (1 - 1)a = q$ so true for $n = 1$

Assume true for $n = k$:

$u_k = q + (k - 1)a$

Prove true for $n = k + 1$:

$u_{k+1} = u_k + a$

$= q + (k - 1)a + a$

$= q + ka - a + a$

$= q + ka$

$= q + (k + 1 - 1)a$

which is $u_n = q + (n - 1)a$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

13) a)

$u_3 = 3u_2 - 2u_1 = 3 \times 13 - 2 \times 5 = 29$

$u_4 = 3u_3 - 2u_2 = 3 \times 29 - 2 \times 13 = 61$

13) b)

Prove true for $n = 1$ and $n = 2$:

$u_1 = -3 + 2^{1+2} = -3 + 8 = 5$ so true for $n = 1$

$u_2 = -3 + 2^{2+2} = 13$ so true for $n = 2$

Assume true for $n = k$:

$u_k = -3 + 2^{k+2}$

Assume true for $n = k + 1$:

$u_{k+1} = -3 + 2^{k+3}$

Prove true for $n = k + 2$:

$u_{k+2} = 3u_{k+1} - 2u_k$

$= 3(-3 + 2^{k+3}) - 2(-3 + 2^{k+2})$

$= -9 + 3 \times 2^{k+3} + 6 - 2 \times 2^{k+2}$

$= -3 + 6 \times 2^{k+2} - 2 \times 2^{k+2}$

$= -3 + 4 \times 2^{k+2}$

$= -3 + 2^{k+4}$

$= -3 + 2^{(k+2)+2}$

which is $u_n = -3 + 2^{n+2}$ with $k + 2$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$